ERICA (KELA) LIU

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EDUCATION

Columbia University Expected Dec 2026
MS in Computer Science
GPA: 3.99 / 4.33

University of California, Santa Barbara Jun 2025
BS in Computer Science, Minor in Philosophy
Dean's Honor | Honors Distinction in Major | GPA: 3.92 / 4.00

EXPERIENCE

Columbia University | Research Assistance Jan 2026 – Present
Multimodal Text-to-Animation Pipeline

- Synthesizing dynamic 2.5D character motion from single static images by integrating depth maps, segmentation labels, and natural language prompts. Aiming for zero visual alteration of the original input photo while achieving accurate character articulation.
- Leveraging image-to-video diffusion models to generate baseline motion priors and extracting skeletal rigs to isolate character movement. Reintegrating the rig back to the image to strictly preserve the fidelity of the background and the character while ensuring temporal motion consistency.

Linx Robot | AI/ML Intern Jun 2025 – Aug 2025
Generative AI Stereo Synthesis Pipeline

- Engineered an end-to-end data generation pipeline using state-of-the-art diffusion models (DDPMs) to accelerate stereo camera R&D.
- Synthesized highly specialized training datasets for downstream CV models, implementing strict conditioning controls to minimize hallucinations and preserve left-eye features for geometry- and pixel-accurate right-eye generation.
- Optimized generative architectures for 2K stereo synthesis under strict compute limits, presenting technical trade-off analyses to drive cross-functional product decisions.

University of California, Irvine | Research Assistance Jul 2024 – Aug 2025
Biomedical Image Segmentation Architecture

- Developed a cascaded 3D U-Net pipeline in PyTorch, designing a sequential learning architecture where each stage leveraged prior predictions (liver → tumor → resection).
- Co-authored an accepted paper for IEEE ISBI 2026 on benchmarking automated 3D segmentation models for liver surgical planning.

University of California, Santa Barbara and FOIA Friend | Capstone Project Sep 2024 – Apr 2025
LLM-Powered Document Evaluation System (FARE)

- Collaborated with start-up company *FOIA Friend* in developing an AI-powered text editor and rating system tailored for FOIA requests. Utilized LLM and Transformers to evaluate FOIA requests and suggest improvements, achieving 82% accuracy on a specifically designed rubric.
- Led integration of rater model into web platform, eliminating need for local download for users. Received over 90% positive feedback from intended users on accessibility and performance.

COURSES AND SKILLS

Relevant Courses: Computer Vision, Computer Graphics, Deep Learning, AI, NLP
Languages: Python, C++, C#, JavaScript
Tools & Frameworks: PyTorch, React.js, Vue.js, Git, Docker, SQL, Unity, Blender